

Sergio Alejandro Rivera Monroy

Civil Engineer | MSc Candidate in Hydraulic Resources Engineering

Specialist in Project Management

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Profile

Civil Engineer with a specialization in Project Management and Integral Project Development from Universidad Escuela Colombiana de Ingeniería Julio Garavito (ECI), currently pursuing a Master's degree in Engineering – Hydraulic Resources at Universidad Nacional de Colombia – Sede Bogotá (UNAL). I have a strong background in academic mentoring for undergraduate students, as well as in providing technical support for the formulation, monitoring, and control of public projects focused on the Bogotá River within the jurisdiction of Cundinamarca, Colombia. Additionally, I have experience conducting hydrological and hydraulic studies for consulting purposes and have carried out university teaching duties in the field of basic hydraulics and fluid mechanics, with solid proficiency in hydrological and hydraulic design software, geographic information systems (GIS), and Microsoft Office. I am recognized for my discipline, organization, and responsibility, with a commitment to continuous improvement. My leadership abilities and critical thinking skills allow me to collaborate effectively in multidisciplinary teams and to take on new challenges and responsibilities with professionalism and dedication.

Professional Experience

Corporación Autónoma Regional de Cundinamarca (CAR)

Civil Engineer

February 2025 – Present

Provision of professional services to the Environmental Infrastructure Technical Directorate (DIA) of CAR Cundinamarca, supporting contracts and agreements in the hydrological and hydraulic components. Conducting field visits to assess the hydrodynamic conditions of river channels and assisting in the technical review of studies related to water resources management.

Universidad Nacional de Colombia – Sede Bogotá (UNAL)

Teaching Assistant

October 2024 – December 2025

Support for teaching activities in the Basic Hydraulics course during the 2024–2 academic term, and in the Fluid Mechanics course during the 2025–1 and 2025–2 academic terms.

SIDESCO S.A.S.

Design Engineer

September 2023 – October 2024

Development of hydrological and hydraulic calculations and modeling for flood hazard, vulnerability, and risk assessments. Analysis of flood extents for photovoltaic park projects and evaluation of scour processes affecting hydrocarbon transportation pipelines. Execution of field campaigns for discharge measurements and sampling of suspended and bed sediments to support sediment transport modeling. Preparation of technical reports, coordination of annexes, and assurance of compliance with the technical scope of each project.

AITEC S.A.S.

Water Resources Engineer

May 2023 – September 2023

Consulting in hydrology, hydraulics, and water quality, including field visits, analysis, and consolidation of results from water resources modeling for multiple contracts. Preparation of technical reports and support for environmental management plan (PMA) oversight processes related to hydro-climatological aspects.

Corporación Autónoma Regional de Cundinamarca (CAR)

Civil Engineer

February 2022 – December 2022

Technical civil support in the formulation and monitoring of projects under the Environmental Investments Fund of the Bogotá River Basin (FIAB). Preparation of quantity takeoff reports, development of AutoCAD 2D drawings, creation of inputs for hydraulic design modeling in Civil 3D, and execution of field visits for

technical support.

Universidad Escuela Colombiana de Ingeniería Julio Garavito (ECI)

Academic Monitor

February 2019 – August 2021

Academic support for students in courses such as Differential Calculus, Linear Algebra, and Geotechnics.

Education

Master of Science (MSc. in Hydraulic Resources)

February 2024 – Expected 2026

Universidad Nacional de Colombia (UNAL) | Bogotá, Colombia

Postgraduate Certificate in Project Management

August 2022 – October 2023

Universidad Escuela Colombiana de Ingeniería Julio Garavito (ECI) | Bogotá, Colombia

Bachelor of Civil Engineering (B. Eng.)

August 2016 – October 2021

Universidad Escuela Colombiana de Ingeniería Julio Garavito (ECI) | Bogotá, Colombia.

Software

Word | Excel | PowerPoint | Microsoft Project | AutoCAD 2D | Civil 3D | ArcGIS Pro | QGIS | HEC – RAS
| EPANET | EPA SWMM | HY – 8 | HEC – HMS | HEC – SSP | TOP MODEL | HBV Light | R | Python

Honors and Awards

- Tuition Waiver Scholarship | 2025 | Universidad Nacional de Colombia – Sede Bogotá
- Graduate Teaching Assistant Fellowship | 2024–2025 | Universidad Nacional de Colombia – Sede Bogotá
- 25 % Partial Graduate Scholarship (Merit-Based) | 2021 | Universidad Escuela Colombiana de Ingeniería Julio Garavito
- National Recognition – Saber Pro Examination (Top Performers) | 2020 | ICFES

Languages

- Spanish: Native
- English: IELTS Academic – Band 6.0 (B2) | February 2026